

Fourier Analysis, Signals & Systems

A Complete Treatment from Intuition to Mathematics

ECE — Signals & Systems

B.Tech First Year

Roadmap

- 1 Signals: The Big Picture
- 2 The Impulse Function & Signals as Weighted Impulses
- 3 Linear Time-Invariant (LTI) Systems & Impulse Response
- 4 Convolution
- 5 Inner Products, Orthogonality & Basis Signals
- 6 Fourier Series
- 7 Fourier Transform
- 8 Fourier Transform of Periodic Signals
- 9 Sampling Theorem
- 10 Discrete-Time Fourier Transform (DTFT)
- 11 Discrete Fourier Transform (DFT)
- 12 Fast Fourier Transform (FFT)
- 13 System Analysis in the Frequency Domain
- 14 Energy & Power Spectral Density
- 15 Multi-Dimensional View: Summary & Connections

What Is a Signal?

Signal

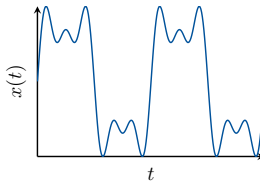
A **signal** is any quantity that varies as a function of one or more independent variables and carries *information*.

$$x : \mathbb{R} \rightarrow \mathbb{R} \quad (\text{or } \mathbb{C})$$

Examples

- Audio: $x(t)$ = air pressure vs. time
- Image: $I(x, y)$ = intensity vs. pixel
- ECG: voltage vs. time
- Radio: EM field amplitude vs. time

A composite signal



Signal Classifications

Property	Types
Time	Continuous / Discrete
Amplitude	Analog / Digital
Periodicity	Periodic / Aperiodic
Energy	Energy / Power signal
Causality	Causal / Non-causal
Symmetry	Even / Odd

Key definitions:

Periodic: $x(t) = x(t + T_0) \forall t$

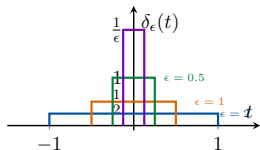
Energy: $E = \int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$

Power: $P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt < \infty$

The Dirac Delta — Intuition

Intuition: A spike of unit area

Imagine squeezing a rectangle of area 1 to zero width. The limit is $\delta(t)$.



Formal definition (distribution):

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\delta(t) = 0 \quad \forall t \neq 0$$

Sifting property:

$$\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0)$$

Properties of $\delta(t)$

Key Properties

$$\delta(at) = \frac{1}{|a|} \delta(t)$$

$$\delta(-t) = \delta(t) \quad (\text{even})$$

$$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$$

$$\frac{d}{dt} u(t) = \delta(t)$$

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

Unit Step

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

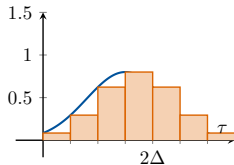
Relationship via derivative

$$\delta(t) = \frac{du(t)}{dt}, \quad u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Any Signal = Weighted Sum of Impulses (CT)

Intuition: Decomposition idea

Approximate $x(t)$ using shifted, scaled rectangles. As width $\rightarrow 0$, each rectangle $\rightarrow$ an impulse.



Riemann sum argument:

$$x(t) \approx \sum_{k=-\infty}^{\infty} x(k\Delta) \delta(t - k\Delta) \cdot \Delta$$

Taking $\Delta \rightarrow 0$:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

This is the **convolution representation** of $x(t)$ with $\delta(t)$!

Signals as Weighted Impulses — Discrete Time

Unit Sample (Discrete δ)

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

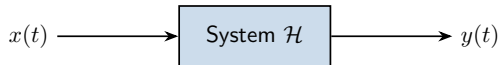
Decomposition of any DT signal:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

Intuition: Why this matters

This identity says $x[n]$ is a *superposition* of scaled, shifted impulses. Because LTI systems are *linear* (respect superposition) and *time-invariant* (shifts commute), knowing how a system responds to a *single* impulse tells us everything about how it responds to *any* input.

Systems: Block Diagram View



Linearity

$$\mathcal{H}\{ax_1 + bx_2\} = a\mathcal{H}\{x_1\} + b\mathcal{H}\{x_2\}$$

Time-Invariance

$$y(t) = \mathcal{H}\{x(t)\} \implies \mathcal{H}\{x(t - t_0)\} = y(t - t_0)$$

Impulse Response $h(t)$

The output when the *input* is $\delta(t)$:

$$h(t) = \mathcal{H}\{\delta(t)\}$$

Causality: $h(t) = 0$ for $t < 0$

Stability (BIBO): $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

Convolution — Where It Comes From

Starting from the decomposition and linearity + time-invariance:

$$\begin{aligned}x(t) &= \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \\y(t) &= \mathcal{H}\{x(t)\} = \mathcal{H}\left\{\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau\right\} \\&= \int_{-\infty}^{\infty} x(\tau) \underbrace{\mathcal{H}\{\delta(t - \tau)\}}_{h(t-\tau)} d\tau\end{aligned}$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

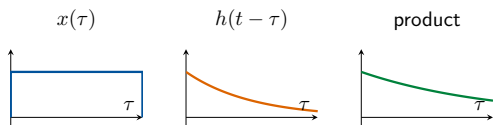
Output of any LTI system = Input convolved with Impulse Response

Convolution — Geometric Interpretation

Intuition: Flip-Slide-Multiply-Integrate

- 1 **Flip** $h(\tau)$ to get $h(-\tau)$
- 2 **Slide** by t : $h(t - \tau)$
- 3 **Multiply** pointwise with $x(\tau)$
- 4 **Integrate** the product over all τ

Repeat for each t to get the entire output signal $y(t)$.



Properties of Convolution

Algebraic Properties

$$x * h = h * x \quad (\text{commutative})$$

$$x * (h_1 + h_2) = x * h_1 + x * h_2 \quad (\text{distributive})$$

$$(x * h_1) * h_2 = x * (h_1 * h_2) \quad (\text{associative})$$

Special Cases

$$x(t) * \delta(t) = x(t)$$

$$x(t) * \delta(t - t_0) = x(t - t_0)$$

$$x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$$

Series/Parallel LTI:

Series: $h = h_1 * h_2$

Parallel: $h = h_1 + h_2$

Discrete-Time Convolution

DT Convolution Sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Derivation mirrors CT:

$$x[n] = \sum_k x[k]\delta[n-k] \implies y[n] = \sum_k x[k]h[n-k]$$

Finite-length signals:

If $x[n]$ has length N and $h[n]$ has length M , then $y[n] = x * h$ has length $N + M - 1$.

All LTI system theory reduces to: **find h , do convolution**. The Fourier/Laplace/Z transforms make this multiplication instead.

Signals as Vectors — Inner Product Spaces

Intuition: Signals live in a vector space

Just as $\mathbf{v} = a_1\hat{e}_1 + a_2\hat{e}_2 + a_3\hat{e}_3$ decomposes a 3D vector, a signal can be decomposed into a (possibly infinite) basis of elementary signals.

Inner Product for CT Signals

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y^*(t) dt$$

Induces norm $\|x\|^2 = \langle x, x \rangle = \int |x(t)|^2 dt$ (energy).

Orthogonality

$x \perp y$ iff $\langle x, y \rangle = 0$. A set $\{\phi_k\}$ is **orthonormal** iff $\langle \phi_k, \phi_l \rangle = \delta_{kl}$.

Projection onto a Basis

Given orthonormal basis $\{\phi_k\}$, project x :

$$x(t) = \sum_k c_k \phi_k(t), \quad c_k = \langle x, \phi_k \rangle = \int x(t) \phi_k^*(t) dt$$

Intuition: Why complex exponentials?

$\phi_k(t) = e^{j\omega_k t}$ are *eigenfunctions* of every LTI system:

$$\mathcal{H}\{e^{j\omega t}\} = H(\omega) e^{j\omega t}$$

where $H(\omega) \in \mathbb{C}$ is just a number (the eigenvalue). This is why sinusoids are the **natural basis** for LTI analysis.

Orthogonality of complex exponentials over $[0, T_0]$:

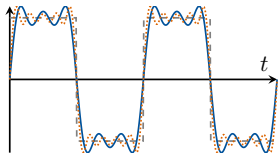
$$\frac{1}{T_0} \int_0^{T_0} e^{jn\omega_0 t} e^{-jm\omega_0 t} dt = \delta_{nm}$$

Fourier Series — Intuition

Intuition: Any periodic signal = sum of harmonically related sinusoids

A guitar string vibrating at 440 Hz also produces 880, 1320, ... Hz (harmonics). The *mixture* determines the timbre. Fourier series makes this precise.

Square wave $\approx$ sum of sines



Observation:

- 1 term: rough approximation
- 3 terms: Closer
- ∞ terms: exact

Key frequencies: $\omega_0, 2\omega_0, 3\omega_0, \dots$

These are **harmonics** of $\omega_0 = 2\pi/T_0$.

Fourier Series — Mathematical Foundation

Complex Exponential Fourier Series

For a T_0 -periodic signal $x(t)$:

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}$$

Synthesis (from coefficients to signal)

Fourier Coefficients (Analysis)

$$c_n = \frac{1}{T_0} \int_{\langle T_0 \rangle} x(t) e^{-jn\omega_0 t} dt$$

$c_n \in \mathbb{C}$; $|c_n|$ = amplitude of n -th harmonic, $\angle c_n$ = phase.

Notation: $\int_{\langle T_0 \rangle}$ means integrate over any interval of length T_0 .

Trigonometric Form of Fourier Series

Using $e^{jn\omega_0 t} = \cos(n\omega_0 t) + j \sin(n\omega_0 t)$:

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$a_0 = \frac{1}{T_0} \int_{\langle T_0 \rangle} x(t) dt \quad (\text{DC component})$$

$$a_n = \frac{2}{T_0} \int_{\langle T_0 \rangle} x(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T_0} \int_{\langle T_0 \rangle} x(t) \sin(n\omega_0 t) dt$$

Relation to complex coefficients: $c_n = \frac{a_n - jb_n}{2}$, $c_0 = a_0$, $c_{-n} = c_n^*$ (for real x)

Worked Example: Square Wave Fourier Series

Signal: $x(t) = \begin{cases} +1 & 0 < t < T_0/2 \\ -1 & T_0/2 < t < T_0 \end{cases}$

Coefficients:

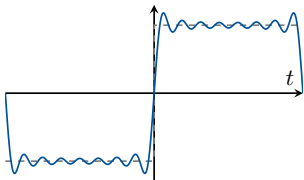
$$\begin{aligned} c_n &= \frac{1}{T_0} \int_0^{T_0/2} e^{-jn\omega_0 t} dt - \frac{1}{T_0} \int_{T_0/2}^{T_0} e^{-jn\omega_0 t} dt \\ &= \frac{1 - e^{-jn\pi}}{jn\pi} = \frac{1 - (-1)^n}{jn\pi} \end{aligned}$$

$$c_n = \begin{cases} \frac{2}{jn\pi} & n \text{ odd} \\ 0 & n \text{ even, } n \neq 0 \\ 0 & n = 0 \end{cases}$$

$$\therefore x(t) = \frac{4}{\pi} \left(\sin \omega_0 t + \frac{\sin 3\omega_0 t}{3} + \frac{\sin 5\omega_0 t}{5} + \dots \right)$$

Convergence & Gibbs Phenomenon

Gibbs phenomenon at discontinuity



- Fourier series converges to the **mean** at discontinuities
- Near jump discontinuities: $\approx 9\%$ **overshoot** persists regardless of N — **Gibbs phenomenon**
- **Dirichlet conditions** guarantee pointwise convergence elsewhere:
 - Finite number of maxima/minima per period
 - Finite number of finite discontinuities
 - $\int_{\langle T \rangle} |x(t)| dt < \infty$

Fourier Series Properties

Property	Signal	Coefficients
Linearity	$ax(t) + by(t)$	$ac_n + bd_n$
Time shift	$x(t - t_0)$	$c_n e^{-jn\omega_0 t_0}$
Frequency shift	$x(t) e^{jm\omega_0 t}$	c_{n-m}
Conjugation	$x^*(t)$	c_{-n}^*
Time reversal	$x(-t)$	c_{-n}
Differentiation	$\frac{dx}{dt}$	$jn\omega_0 c_n$
Integration	$\int_{-\infty}^t x d\tau$	$\frac{c_n}{jn\omega_0}, c_0 = 0$
Convolution (period)	$x \circledast y$	$T_0 c_n d_n$
Multiplication	$x(t) y(t)$	$\sum_k c_k d_{n-k}$

Parseval's Theorem for Fourier Series

Parseval's Relation (Power)

$$P = \frac{1}{T_0} \int_{\langle T_0 \rangle} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |c_n|^2$$

Intuition: Energy conservation in frequency domain

Total average power equals the sum of powers at each harmonic. The spectrum $\{|c_n|^2\}$ is the **power spectrum**.

Example application: For the square wave,

$$P = 1, \quad \sum_{\substack{n=-\infty \\ n \text{ odd}}}^{\infty} \frac{4}{n^2 \pi^2} = \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{8}{\pi^2} \cdot \frac{\pi^2}{8} = 1 \checkmark$$

From Fourier Series to Fourier Transform

Intuition: What happens as $T_0 \rightarrow \infty$?

A periodic signal with $T_0 \rightarrow \infty$ becomes *aperiodic*. The discrete frequencies $n\omega_0$ become *continuous* ω . The spectrum c_n becomes a *density* $X(\omega)$.

Start: $x(t) = \sum_n c_n e^{jn\omega_0 t}$, $c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt$

Define $X(n\omega_0) = T_0 c_n = \int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt$

As $T_0 \rightarrow \infty$: $\omega_0 = \frac{2\pi}{T_0} \rightarrow d\omega$, **sum** $\rightarrow$ **integral**:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega, \quad X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

The Fourier Transform Pair

Fourier Transform (Analysis)

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$X(\omega) \in \mathbb{C}$: encodes amplitude and phase at each frequency ω (rad/s)

Inverse Fourier Transform (Synthesis)

$$x(t) = \mathcal{F}^{-1}\{X(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Alternative (f -convention):

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt, \quad x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

(symmetric — no 2π factors; common in EE textbooks)

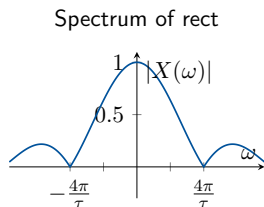
Fourier Transform — Worked Examples

1. Rectangular pulse:

$$x(t) = \text{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1 & |t| < \tau/2 \\ 0 & \text{else} \end{cases}$$

$$X(\omega) = \int_{-\tau/2}^{\tau/2} e^{-j\omega t} dt = \tau \text{sinc}\left(\frac{\omega\tau}{2\pi}\right)$$

where $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$.



More Fourier Transform Pairs

$x(t)$	$X(\omega)$
$\delta(t)$	1
1	$2\pi \delta(\omega)$
$e^{-at}u(t), a > 0$	$\frac{1}{a + j\omega}$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$u(t)$	$\pi \delta(\omega) + \frac{1}{j\omega}$
$\text{rect}(t/\tau)$	$\tau \text{sinc}(\omega\tau/2\pi)$
$\text{sinc}(Wt)$	$\frac{\pi}{W} \text{rect}\left(\frac{\omega}{2W}\right)$

Fourier Transform Properties

Property	$x(t)$	$X(\omega)$
Linearity	$ax + by$	$aX + bY$
Time shift	$x(t - t_0)$	$e^{-j\omega t_0} X(\omega)$
Frequency shift	$e^{j\omega_0 t} x(t)$	$X(\omega - \omega_0)$
Time scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{\omega}{a}\right)$
Time reversal	$x(-t)$	$X(-\omega) = X^*(\omega)$ [real x]
Differentiation	$\frac{d^n x}{dt^n}$	$(j\omega)^n X(\omega)$
Integration	$\int_{-\infty}^t x \, d\tau$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Convolution	$x * h$	$X(\omega) \cdot H(\omega)$
Multiplication	$x(t) \cdot y(t)$	$\frac{1}{2\pi} X * Y$
Duality	$X(t)$	$2\pi x(-\omega)$

The Convolution Theorem — Key Result

Convolution Theorem

$$y(t) = x(t) * h(t) \longleftrightarrow Y(\omega) = X(\omega) \cdot H(\omega)$$

Convolution in time = pointwise multiplication in frequency.

Intuition: Why this is powerful

- Computing convolution directly: $O(N^2)$
- Via FFT: $O(N \log N)$ (factor of thousands faster for large N)
- System analysis: just divide $Y(\omega)$ by $X(\omega)$

Frequency Response of LTI system:

$$H(\omega) = \mathcal{F}\{h(t)\} = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$

$|H(\omega)|$ = **magnitude response**, $\angle H(\omega)$ = **phase response**

Parseval's Theorem for Fourier Transform

Parseval / Plancherel Relation (Energy)

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$\frac{1}{2\pi} |X(\omega)|^2 = S(\omega) \quad (\text{Energy Spectral Density})$$

Proof sketch:

$$\begin{aligned} \int |x|^2 dt &= \int x(t) x^*(t) dt = \int x(t) \left[\frac{1}{2\pi} \int X^*(\omega) e^{-j\omega t} d\omega \right] dt \\ &= \frac{1}{2\pi} \int X^*(\omega) \underbrace{\int x(t) e^{-j\omega t} dt}_{X(\omega)} d\omega = \frac{1}{2\pi} \int |X(\omega)|^2 d\omega \end{aligned}$$

Uncertainty Principle

Time-Bandwidth Uncertainty

$$\sigma_t \sigma_\omega \geq \frac{1}{2}$$

$$\text{where } \sigma_t^2 = \frac{\int t^2 |x|^2 dt}{\int |x|^2 dt}, \quad \sigma_\omega^2 = \frac{\int \omega^2 |X|^2 d\omega}{2\pi \int |x|^2 dt}$$

Intuition: Narrower pulse = wider spectrum

A short radar pulse has good time resolution but poor frequency resolution. A long pulse has good frequency resolution but poor time resolution. **Trade-off is fundamental, not a limitation of technology.**

Equality holds for Gaussian: $x(t) = e^{-t^2/2\sigma^2}$, $X(\omega) = \sigma\sqrt{2\pi} e^{-\sigma^2\omega^2/2}$

Fourier Transform of Periodic Signals

Issue: Periodic signals are *not* absolutely integrable $\Rightarrow$ FT doesn't converge in the classical sense.

Solution: Use distributions (Dirac deltas).

FT of Periodic Signal

If $x(t) = \sum_n c_n e^{jn\omega_0 t}$, then:

$$X(\omega) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0)$$

- Spectrum = impulse train in frequency domain
- Impulse at $n\omega_0$ has weight $2\pi c_n$
- Neatly connects Fourier series $\leftrightarrow$ Fourier transform

Sampling — Intuition

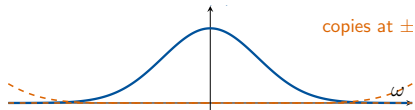
Intuition: How do we store a continuous signal digitally?

Sample it periodically at rate f_s . The question: what is the *minimum* sampling rate to avoid losing information?

Sampling model: $x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_n x(nT_s) \delta(t - nT_s)$

FT of sampled signal:

$$X_s(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s), \quad \omega_s = \frac{2\pi}{T_s}$$



Nyquist-Shannon Sampling Theorem

Nyquist-Shannon Theorem

A band-limited signal with $X(\omega) = 0$ for $|\omega| > \omega_{\max}$ can be **perfectly reconstructed** from its samples taken at rate

$$\omega_s \geq 2\omega_{\max} \quad \Leftrightarrow \quad f_s \geq 2f_{\max}$$

The minimum rate $f_s = 2f_{\max}$ is the **Nyquist rate**.

Reconstruction (ideal low-pass filter with cutoff $\omega_s/2$):

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

- **Aliasing:** if $f_s < 2f_{\max}$, spectral copies overlap — frequencies are indistinguishable
- **Anti-aliasing filter:** low-pass filter before ADC

DTFT — Moving to Discrete Time

Intuition: Continuous $\rightarrow$ Discrete

Replace $x(t)$ with $x[n]$ (sequence of samples). Continuous frequency $\omega \rightarrow$ normalized discrete frequency $\hat{\omega} \in [-\pi, \pi]$.

DTFT Pair

$$X(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\hat{\omega}n} \quad (\text{Analysis})$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\hat{\omega}}) e^{j\hat{\omega}n} d\hat{\omega} \quad (\text{Synthesis})$$

Key properties:

- $X(e^{j\hat{\omega}})$ is **always** 2π -periodic in $\hat{\omega}$
- For real $x[n]$: $X(e^{-j\hat{\omega}}) = X^*(e^{j\hat{\omega}})$ (conjugate symmetry)
- Normalized frequency: $\hat{\omega} = \omega T_s$ (rad/sample)

DTFT Properties & Examples

Convolution theorem (DT):

$$y[n] = x * h \leftrightarrow Y = X \cdot H$$

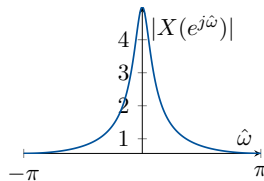
Parseval (DT):

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\hat{\omega}})|^2 d\hat{\omega}$$

Example: $x[n] = a^n u[n]$, $|a| < 1$:

$$X(e^{j\hat{\omega}}) = \frac{1}{1 - ae^{-j\hat{\omega}}}$$

$|X|$ for $a = 0.8$



DFT — Why We Need It

- Fourier Series: periodic & continuous $\rightarrow$ discrete coefficients
- CTFT: aperiodic & continuous $\rightarrow$ continuous spectrum
- DTFT: discrete $\rightarrow$ continuous, 2π -periodic spectrum
- **DFT: discrete input, discrete spectrum** — *computable!*

Intuition: Core idea

Take N samples of a signal. Assume it's periodic with period N . Compute N frequency samples of the DTFT.



DFT Definition

N -Point DFT Pair

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad k = 0, 1, \dots, N-1 \quad (\text{Analysis})$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad n = 0, 1, \dots, N-1 \quad (\text{Synthesis})$$

where the **twiddle factor** is:

$$W_N = e^{-j2\pi/N} \quad (\text{primitive } N\text{-th root of unity})$$

Physical meaning of k : Frequency bin k corresponds to analog frequency $f_k = \frac{k f_s}{N}$, i.e., $\omega_k = \frac{2\pi k}{N}$ rad/sample.

DFT as a Matrix Operation

DFT Matrix

$\mathbf{X} = \mathbf{W}_N \mathbf{x}$, where $\mathbf{W}_N \in \mathbb{C}^{N \times N}$:

$$[\mathbf{W}_N]_{k,n} = W_N^{kn} = e^{-j2\pi kn/N}$$

For $N = 4$:

$$\mathbf{W}_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix}$$

Key: $\mathbf{W}_N$ is **unitary** (up to $\sqrt{N}$): $\mathbf{W}_N \mathbf{W}_N^H = N\mathbf{I}$

Hence $\mathbf{W}_N^{-1} = \frac{1}{N} \mathbf{W}_N^H$.

DFT Properties

Property	$x[n]$	$X[k]$
Linearity	$ax + by$	$aX + bY$
Circular shift	$x[\langle n - m \rangle_N]$	$W_N^{km} X[k]$
Frequency shift	$W_N^{-\ell n} x[n]$	$X[\langle k - \ell \rangle_N]$
Circular convolution	$x \circledast y$	$X[k] \cdot Y[k]$
Multiplication	$x[n] \cdot y[n]$	$\frac{1}{N} X \circledast Y$
Parseval	$\sum x[n] ^2$	$\frac{1}{N} \sum X[k] ^2$
Conjugate sym. (real x)	$x[n] \in \mathbb{R}$	$X[N - k] = X^*[k]$

Circular convolution: $y[n] = (x \circledast h)[n] = \sum_{m=0}^{N-1} x[m] h[\langle n - m \rangle_N]$

To use DFT for *linear* convolution: **zero-pad** to length $\geq N + M - 1$.

Worked DFT Example

$N = 4$, $x = [1, 2, 3, 4]^T$. Compute $X[k]$:

$$X[0] = 1 + 2 + 3 + 4 = 10$$

$$X[1] = 1 + 2(-j) + 3(-1) + 4(j) = (1 - 3) + j(-2 + 4) = -2 + 2j$$

$$X[2] = 1 + 2(-1) + 3(1) + 4(-1) = 1 - 2 + 3 - 4 = -2$$

$$X[3] = 1 + 2(j) + 3(-1) + 4(-j) = -2 - 2j$$

Note: $X[3] = X^*[1]$ — conjugate symmetry for real input.
Only $X[0], X[1], X[2]$ (first $\lfloor N/2 \rfloor + 1$ bins) are independent.

FFT — The Most Important Algorithm in Engineering

Intuition: Problem

Direct DFT: $O(N^2)$ multiplications. For $N = 10^6$: too slow.

FFT: $O(N \log_2 N)$ — 10^6 samples: 10^6 vs 2×10^7 operations.

Cooley-Tukey Radix-2 DIT (Decimation-In-Time):

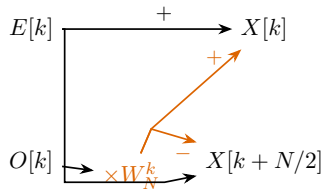
Split $x[n]$ into even/odd indexed sequences (N must be power of 2):

$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn} \\ &= \sum_{m=0}^{N/2-1} x[2m] W_N^{2mk} + \sum_{m=0}^{N/2-1} x[2m+1] W_N^{(2m+1)k} \\ &= \underbrace{\sum_{m=0}^{N/2-1} x[2m] W_{N/2}^{mk}}_{E[k]} + W_N^k \underbrace{\sum_{m=0}^{N/2-1} x[2m+1] W_{N/2}^{mk}}_{O[k]} \end{aligned}$$

FFT Butterfly Structure

Butterfly computation (core FFT operation):

$$X[k] = E[k] + W_N^k O[k], \quad X[k + N/2] = E[k] - W_N^k O[k]$$



Apply recursively: N -point FFT $\rightarrow$ two $N/2$ -point FFTs $\rightarrow \dots \rightarrow \log_2 N$ stages, each $N/2$ butterflies $\rightarrow \frac{N}{2} \log_2 N$ multiplications.

Transfer Function & Frequency Response

Transfer Function

For an LTI system with input $X(\omega)$ and output $Y(\omega)$:

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

Magnitude response $|H(\omega)|$:

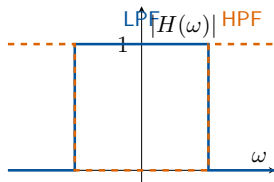
- Scales each frequency component
- Low-pass: passes low ω , blocks high
- High-pass: vice versa
- Band-pass: passes a band

Phase response $\angle H(\omega)$:

- Phase shift at each frequency
- **Linear phase** $\Rightarrow$ pure time delay (no distortion)
- Non-linear phase $\Rightarrow$ phase distortion

Ideal Filters

Ideal Filter Types



Ideal LPF:

$$H(\omega) = \begin{cases} e^{-j\omega t_d} & |\omega| \leq \omega_c \\ 0 & \text{else} \end{cases}$$

$$\Rightarrow h(t) = \frac{\omega_c}{\pi} \text{sinc}\left(\frac{\omega_c(t - t_d)}{\pi}\right)$$

Problem: $h(t) \neq 0$ for $t < 0 \Rightarrow$ **non-causal** (unrealizable). Practical filters approximate this.

Distortionless Transmission

Distortionless System

$$y(t) = K x(t - t_d) \text{ for constants } K, t_d.$$

In frequency domain:

$$Y(\omega) = K e^{-j\omega t_d} X(\omega) \implies H(\omega) = K e^{-j\omega t_d}$$

Requirements:

- $|H(\omega)| = K$ (constant magnitude — flat spectrum)
- $\angle H(\omega) = -\omega t_d$ (linear phase)

Group delay:

$$\tau_g(\omega) = -\frac{d}{d\omega} \angle H(\omega)$$

Constant group delay $\Leftrightarrow$ linear phase $\Leftrightarrow$ distortionless.

LCCDE and Frequency Response

An N -th order LTI system is described by:

$$\sum_{k=0}^N a_k \frac{d^k y}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x}{dt^k}$$

Apply FT (using differentiation property $(j\omega)^k$):

$$\left(\sum_{k=0}^N a_k (j\omega)^k \right) Y(\omega) = \left(\sum_{k=0}^M b_k (j\omega)^k \right) X(\omega)$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} = \frac{B(j\omega)}{A(j\omega)}$$

Poles ($A(j\omega) = 0$): resonances, instability risk

Zeros ($B(j\omega) = 0$): signal nulls at those frequencies

Bode Plot Sketching

Intuition: Log-frequency representation

$|H(j\omega)|_{\text{dB}} = 20 \log_{10} |H(j\omega)|$; plot vs $\log_{10} \omega$. Asymptotes are easy to sketch.

Standard factors and their Bode plots:

Factor	Magnitude	Phase
K	$20 \log K $ dB (flat)	0° or 180°
$(j\omega)^{\pm n}$	$\pm 20n$ dB/dec	$\pm 90n^\circ$
$(1 + j\omega/\omega_c)^{\pm 1}$	± 20 dB/dec above ω_c	$\pm 45^\circ$ at ω_c
Underdamped 2nd order	± 40 dB/dec; resonance peak	$0 \rightarrow \pm 180^\circ$

Spectral Density

Energy Spectral Density (ESD)

$$S_{xx}(\omega) = |X(\omega)|^2$$

Energy in band $[\omega_1, \omega_2]$: $E_{12} = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} S_{xx}(\omega) d\omega$

Power Spectral Density (PSD)

$$\mathcal{S}_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T}$$

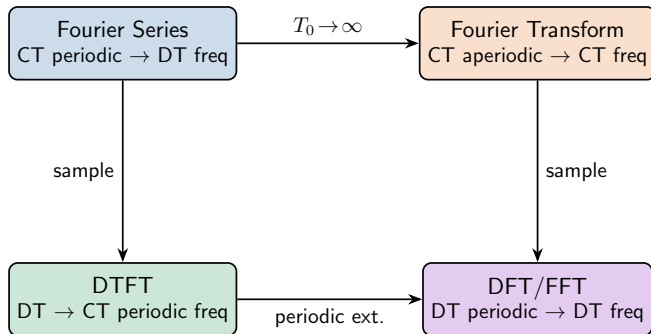
where $X_T(\omega) = \int_{-T/2}^{T/2} x(t) e^{-j\omega t} dt$.

Wiener-Khinchin theorem:

$$\mathcal{S}_{xx}(\omega) = \mathcal{F}\{R_{xx}(\tau)\}, \quad R_{xx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t+\tau) dt$$

PSD is the FT of the **autocorrelation function**.

The Fourier Family — Unified View



Summary: Key Formulas

Transform	Forward	Inverse
FS	$c_n = \frac{1}{T_0} \int_{\langle T_0 \rangle} x(t) e^{-jn\omega_0 t} dt$	$x(t) = \sum_n c_n e^{jn\omega_0 t}$
FT	$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$
DTFT	$X(e^{j\hat{\omega}}) = \sum_n x[n] e^{-j\hat{\omega} n}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\hat{\omega}}) e^{j\hat{\omega} n} d\hat{\omega}$
DFT	$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$	$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$

Convolution theorem holds for ALL of the above:

$$y = x * h \longleftrightarrow Y = X \cdot H$$

Summary: Physical Meanings

- ❶ **Impulse** $\delta(t)$: unit-area spike; sifting property extracts values.
- ❷ **Signals as impulses**: $x(t) = \int x(\tau)\delta(t - \tau)d\tau$ — superposition.
- ❸ **Impulse response** $h(t)$: complete characterization of any LTI system.
- ❹ **Convolution**: $y = x * h$ — output of LTI to any input.
- ❺ **Fourier Series**: periodic signal $\leftrightarrow$ discrete spectrum $\{c_n\}$.
- ❻ **Fourier Transform**: aperiodic signal $\leftrightarrow$ continuous spectrum $X(\omega)$.
- ❼ **Convolution theorem**: multiplication in frequency = convolution in time.
- ❽ **DTFT**: FT for sampled signals; spectrum is 2π -periodic.
- ❾ **DFT**: computable frequency analysis for finite sequences.
- ❿ **FFT**: $O(N \log N)$ algorithm for DFT — backbone of DSP.
- ⓫ **Frequency response** $H(\omega)$: tells how a system filters each frequency.

Further Study: What Comes Next

Immediate next topics:

- Laplace Transform (generalizes FT, handles unstable systems)
- Z-Transform (DT analog of Laplace)
- Poles & zeros, stability criteria
- Filter design (Butterworth, Chebyshev, FIR/IIR)

Applications you can now understand:

- AM/FM modulation (frequency shift)
- Audio equalization (frequency shaping)
- Image compression (2D DFT/DCT in JPEG)
- SDR signal processing
- OFDM (FFT-based modulation in WiFi/LTE)
- Radar pulse compression (matched filtering)

The End — You Now Speak Frequency Domain.